21000563 Experimental Design and Analysis of Variance
AY 2026 - Semester 1

Project Final Report

Class 1
Team 5

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1. Scientific Question

1.1 Practical Motivation

As cut flowers are primarily traded for special occasions, it is important to maintain their ornamental value for a long time. Vase life is perceived as an important quality indicator that influences marketability and consumer satisfaction (Sutrisno et al., 2025). It is also a significant factor in repurchase (Vehniwal & Abbey, 2019). Therefore, previous studies have examined preservation methods for extending the vase life of cut flowers, including simple treatments that can be applied by everyday consumers.

As such, while various methods for maintaining flower freshness have been reported in the literature, these guidelines often cover general information rather than methods tailored to specific flower types. However, each flower possesses unique physiological characteristics, such as water absorption rates and temperature sensitivity. Therefore, it is meaningful to select a specific type of flower and research methods to extend its vase life.

In the survey of cut flower distribution in the private market from May to October 2022, rose sales outlets were the most numerous, except for May (Lee et al., 2022). This suggests that roses are the most easily accessible flowers for consumers. Since roses are among the most commonly purchased cut flowers, identifying simple preservation methods for extending their vase life has practical relevance for general consumers. Rather than focusing on professional floristry techniques, this study examines preservation conditions that can be easily applied by the general public.

1.2 Research Question

Among commonly used flower preservation methods, this study investigates the effect of adding sugar and vinegar to water on the vase life of cut roses. The research question is to determine the water treatment method that most effectively preserves cut roses among tap water, sugar water, vinegar water, and sugar-vinegar water.

1.3 Validity of the Experimental Approach

In this experiment, all roses were purchased on the same day from the same market and screened for visible damage prior to assignment, minimising pre-existing differences between experimental units. Stems were trimmed to a uniform length and thorns removed under identical conditions, ensuring that variation in initial state did not confound the treatment comparisons.

Vase life was assessed using the VBN rose evaluation card, a standardised protocol developed specifically for cut roses and widely used in postharvest research. This ensured that termination decisions were consistent and reproducible across both locations and observers, so that the recorded vase life reflects the effect of the treatment rather than differences in how termination was judged.

On this bias, these measures support the claim that systematic differences in vase life among the four treatment combinations can be attributed to the additives under investigation rather than to extraneous sources of variation.

2. Scope

The experiment was conducted as a blocked 2×2 factorial design to examine how two vase-solution additives affect the vase life of cut roses. The two treatment factors were Sugar (with or without) and Vinegar (with or without), resulting in four treatment combinations. The experimental unit was a single cut rose placed individually in a vase, treated and observed independently throughout the experiment. Based on the a priori power analysis described in Section 7, eight replications were assigned to each treatment combination.

Because available space was limited, the 32 experimental units were divided between two physical locations. Location was treated as a blocking factor to account for nuisance variability arising from the two locations being separate households with distinct environmental conditions. Each location contained four replications of every treatment combination, preserving a balanced factorial structure within each block.

Within each block, the physical positions of the vases were randomised in R using a reproducible randomisation procedure. This design therefore allowed the effects of Sugar, Vinegar, and their interaction to be examined with location-related variability absorbed as a block effect.

3. Treatments and Response Variable

3.1 Justification for Treatments

3.1.1 Water Type as a Factor

Methods used to extend the vase life of cut flowers and maintain freshness include cold storage and wet storage with cut flower preservative (KATI, 2007). Since cold storage is a method used to maintain freshness during distribution, it was considered appropriate to discuss wet storage with cut flower preservatives rather than cold storage.

3.1.2 Sugar and Vinegar as Treatment Factors

Sucrose, 8-HQS (8-hydroxyquinoline sulfate), STS (silver thiosulfate), organic acids, and antioxidants are commonly used to extend the vase life and maintain the freshness of cut flowers (KATI, 2007). Sugar acts as respiratory substrates to delay aging, increase body weight, and facilitate water absorption by raising osmotic pressure. Aluminum sulfate, HQC, HQS, etc., exert a bactericidal effect and facilitate water absorption by acidifying the preservative solution (Kim, 2001). Therefore, among readily available items in daily life, white sugar was selected from sugars that delay aging, and white vinegar was selected from acidic substances that acidify the preservative solution. This choice is further supported by De Silva et al. (2013), who used sucrose- and vinegar-based preservative solutions.

3.2 Definition of Treatments

Tap water, 2% Sugar water, 0.6% White Vinegar water, 2% Sugar + 0.6% White Vinegar water is used in the experiment. To prevent differences between the initial and subsequent water temperature, each treatment is kept at room temperature for 30 minutes before use. To make the experiment easy to conduct at home, the most readily available Beksul White Sugar and OTOKI White Vinegar are used for this experiment. White Vinegar is a colorless, transparent vinegar made using distilled spirits produced by fermenting grains. Unlike regular vinegar, it contains no sugar. Therefore, it was selected to examine only the effects of pure vinegar.

The first treatment is tap water, which serves as the control in this experiment. It is defined as 120ml of tap water left at room temperature for 30 minutes.

The second treatment is sugar water, prepared at a concentration of 2%. 10 g of sugar is dissolved in tap water, and the final volume is adjusted to 500mL. After dissolution, the solution is left at room temperature for 30 minutes. Then, 120mL of the solution was dispensed into each vase. The 2% concentration is adopted from *De Silva et al.* (2013).

The third treatment is white vinegar water, prepared at a concentration of 0.6%. As measuring 0.6 mL directly poses practical difficulties, 3 mL of white vinegar is added to tap water and the final volume adjusted to 500 mL. The solution is left at room temperature for 30 minutes. Then, put 120mL of the solution into each vase. The 0.6% concentration is adopted from De Silva et al. (2013).

The fourth treatment is a solution of 2% sugar and 0.6% white vinegar. 3 mL of white vinegar and 10 g of sugar is added to tap water, and the final volume is adjusted to 500 mL. The solution is left at room temperature for 30 minutes. Then, put 120mL of the solution into each vase. The treatment concentration is adopted from De Silva et al. (2013).

To minimize the risk of microbial contamination, each treatment is replaced at 48-hour intervals throughout the experimental period.

3.3 Justification for Response Variable

3.3.1 Reason for selecting Vase life as the response variable

Quality evaluation in the cut flower industry is generally based on external characteristics such as color and shape(In et al., 2016). Although vase life and freshness are not included in grading criteria, these are also important factors in consumers' flower selection (In et al., 2016). Short and unpredictable vase life leads to a decline in customer satisfaction (In et al., 2016). Vase life is an important quality indicator that influences marketability and consumer satisfaction (Sutrisno et al., 2025). Vase life is not only one of the most important factors influencing customer satisfaction, but also a significant factor in repurchase (Vehniwal & Abbey, 2019). Therefore, this study investigates methods to sustain the ornamental value of roses by focusing on vase life rather than external grading factors.

3.3.2 Reason for measuring in 12-hour increments

In this experiment, changes in the quality of cut roses are observed at 12-hour intervals. This interval was established considering the characteristics of roses, which undergo rapid physiological changes after harvest (KATI, 2007). Observations at 12-hour intervals are commonly observed in cut flower research (Sui et al., 2015).

3.3.3 Justification for Termination Criteria

VBN card provides standardised evaluation criteria developed specifically for cut roses. Furthermore, the Netherlands accounts for 60% of the global flower market (KATI, 2023). As VBN is the higher-level organization overseeing Dutch cooperative flower auctions, VBN standards can be considered a globally reliable indicator.

3.4 Precise Definition of Response Variable

The response variable is rose freshness, defined as the number of days from placement in the vase until any of the termination criteria specified by the VBN Rosa Evaluation Card (FloraHolland, 2014) are met.

3.4.1 Termination Criteria : VBN Rosa Evaluation Card

Termination criteria were established using the VBN Rosa Evaluation Card (FloraHolland, 2014), which was developed by VBN (Vereniging van Bloemenveilingen in Nederland), the leading Dutch flower auction association and a widely recognised authority in postharvest quality standards for cut flowers. This card provides standardised evaluation criteria developed specifically for cut roses.

The VBN card encompasses a wide range of criteria, but applying all of them was not practical given the time constraints and the risk of subjective judgment. Six criteria were therefore chosen: those with clear, observable write-off conditions to minimise ambiguity, and those covering different parts of the flower so that both floral and foliar deterioration could be considered.

Table 3.4.1.1 *Flower-Related Termination Criteria*

Criterion	Description	Write-off
Bent Neck	Drying and kinking of the stem just below the bud	Flower angle > 90°
Brown Flower	Petals clearly turning brown, not due to Botrytis	> 50% of total petal surface area
End of Flowering	Rose fully opened (stage 4–5), petals falling and/or discoloring	> 2 petals fallen, or flower visibly limp *(discoloration alone NOT sufficient)*

Bent neck was selected as criterion because it represents a well-recognised indicator of vascular deterioration in cut roses (Daneshmand et al., 2024).

Brown Flower and End of Flowering were additionally included as both target distinct aspects of flower deterioration and provide relatively clear and observable indicators for visual assessment.

Table 3.4.1.2 *Stem-Related Termination Criteria*

Criterion	Description	Write-off
Kinking	The stem kinks	Upon detection
Botrytis	The stem discolors gray to black, sometimes gray fluffy mold appears, the part of the above the affected area withers and the flower withers.	Upon detection

For stem-related criteria, both kinking and botrytis were included. Kinking was selected as a direct indicator of stem condition. Botrytis was retained as one of the most frequently occurring fungal pathogens in cut roses (Muñoz et al., 2019)

Table 3.4.1.3 *Foliage-Related Termination Criteria*

Criterion	Description	Write-off
Yellow and/or Brown Foliage	Leaves turn yellow and/or brown	> 50% of total leaf surface area or > 50% of total leaf count

Yellow and/or Brown Foliage accounts for visual deterioration, and this covers qualitative aspects of foliage condition.

3.4.2 Measurement Method

Observations are conducted twice daily at 08:00 and 20:00, providing a consistent 12-hour interval. Freshness is recorded in 0.5-day increments for each session. Visual scoring follows the termination criteria specified in Section 3.4.1. Each vase is assessed as a single experimental unit.

All observers use a Double A Company's A4 white paper sheet as a shared colour reference background during assessment. All observations are conducted under artificial lighting to eliminate variability caused by natural light fluctuations throughout the day.

Environmental conditions, specifically temperature and humidity, are monitored throughout the experiment. Both factors are known to influence postharvest quality and the rate of floral deterioration in cut roses (Daneshmand et al., 2024).

Visual logs are recorded photographically at each observation session using a smartphone mounted on a fixed tripod at a consistent position. To maximise colour accuracy and reproducibility, the camera is set to manual mode with the following fixed parameters.

- ISO 50 : To minimise image noise
- White balance 5500K : To approximate natural daylight and ensure colour consistency across sessions
- Shutter speed 1/10s : To eliminate flickering artefacts from artificial lighting
- In-built software colour correction disabled : To prevent automatic adjustments from altering colour representation between sessions

3.4.3 Supplementary Measurement : ImageJ

Relying on visual observation alone involved a risk of subjectivity, especially when evaluating petal discoloration, because subtle colour changes could be perceived differently by observers. Therefore, ImageJ, a widely used open-source image analysis software for quantifying colour changes and morphological features, was used as a supplementary measurement tool (ImageJ, 2007).

During each observation session, the termination criteria were first assessed visually using the VBN card. When visual inspection suggested that more than 50% of the petals or foliage met the termination criteria, the corresponding photographs were then checked using ImageJ to support the assessment.

Captured image records were adjusted using an A4 white sheet of paper as a white reference, improving consistency in colour assessment despite variation in lighting across sessions and locations. ImageJ was then used to extract quantitative colour information from the relevant petal or foliage regions. This procedure improved the objectivity of discolouration assessment and reduced reliance on visual judgement alone.

Representative image records and analysis procedure for flowers terminated due to petal discolouration or foliage-related criteria are provided in Appendix A.

4. Treatment Assignment

4.1 Blocking by Location

As noted in §2, location was treated as a blocking factor. Since the two locations were separate households, they were expected to differ in environmental conditions such as light exposure, air circulation, and room temperature. Location was therefore treated as a blocking factor to account for this nuisance variability. Each location contained a complete and balanced set of the factorial design, with four replications of each treatment combination.

4.2 Treatment Labelling and Position Randomisation

Within each location, 16 vases were used, corresponding to four replications of each of the four treatment combinations. For logistical consistency in solution preparation, vases were numbered 1–16 and assigned to treatments in fixed blocks: vases 1–4 received Tap water, 5–8 received Sugar, 9–12 received Vinegar, and 13–16 received Sugar + Vinegar. This labelling was fixed for preparation purposes and was not itself a randomisation step.

4.2.1 Vase Position Randomisation

To prevent this fixed vase-to-treatment labelling from being confounded with the physical environment, the experimental area within each location was divided into 16 designated positions (each corresponding to half of an A4 sheet). Vase numbers 1–16 were then randomly assigned to these positions in R using a fixed seed (set.seed(99)), ensuring that the physical placement of each treatment was randomised within each location.

Stems were screened prior to assignment: those with already discolored or torn petals, or with grayish stems, were excluded. Eligible stems were then inserted into the assigned vases without additional sorting or deliberate selection based on visible stem characteristics.

Table 4.2.1.1 *Final Randomised Vase Position Assignment*

Position	**Vase**	**Position**	**Vase**
1	16	**9**	8
2	1	**10**	14
3	12	**11**	11
4	6	**12**	4
5	13	**13**	7
6	5	**14**	10
7	3	**15**	9
8	2	**16**	15

5. Nuisance Variation

5.1 Plant Material Variability

5.1.1 Stem Cutting Method

All stems were recut to a length of 18–20 cm before treatment application. This standardisation reduced variability in stem length and helped ensure comparable exposure to the vase solution across experimental units. This practice is consistent with prior findings that recutting amount and stem length can influence vase life in cut roses (Moody, Dole, and Barnes, 2014).

5.1.2 Individual Stem Variability

Individual stems may differ in physiological condition, such as bud maturity or vascular health, even when they originate from the same wholesale batch. Because these differences could not be fully observed or controlled in advance, stems were assigned to treatments without selection based on individual characteristics. The detailed randomisation procedure, including stem pre-screening and insertion order, is described in Section 4. This procedure was intended to distribute stem-level variability across treatment groups in expectation, thereby reducing the risk that unobserved stem differences would bias treatment effect estimates.

5.2 Environmental and Spatial Variability

5.2.1 Temperature, Humidity, and Microbial Contamination

Previous postharvest experiments have maintained strict environmental control, including constant temperature and uniform light intensity (De Silva et al., 2013). In the present study, however, complete environmental control was not feasible because the experiment was conducted in a home setting.

Temperature and humidity were therefore monitored and recorded with a digital thermo-hygrometer at 08:00 and 20:00 each day. The experiment was conducted indoors at an ambient temperature of approximately 25–28°C, which was higher than the 20°C condition maintained by De Silva et al. (2013). Since higher ambient temperatures can accelerate bacterial proliferation in vase solutions (Van Doorn et al., 2013), regular solution replacement was necessary. However, replacing sugar-based solutions too frequently could continuously supply fresh nutrients for microbial growth (De Silva et al., 2013). To balance these concerns and to keep handling consistent across treatments, all vase solutions were replaced every two days.

5.2.2 Location Blocking

The two locations were separate households and were therefore expected to differ in light exposure, air circulation, and other site-specific conditions. As described in Section 4.1, location was treated as a blocking factor. Because all treatment combinations were represented within each location, location effects were not confounded with the treatment main effects and could be included as a blocking factor in the analysis.

5.2.3 Light Exposure Within Location

Within each location, exposure to natural light could have varied depending on a unit's position relative to windows. To reduce this source of variability, curtains were drawn throughout the experiment to block natural light, and each location was illuminated only under the artificial lighting condition also used during measurement.

5.2.4 Vase and Container Uniformity

All experimental units used the same vase model (Daiso, item no. 1052134) to minimise container-related differences in volume, shape, and material.

5.3 Measurement and Observation Variability

5.3.1 Observer Subjectivity

Two observers participated in the observation process, with each observer responsible for one location throughout the experiment. Because each observer was responsible for one location, any remaining observer-specific differences could not be separated from location effects. However, all treatment combinations were balanced within each location, so these observer-location differences were not systematically aligned with any treatment combination. Observer subjectivity was further reduced by applying a standardized VBN protocol, documenting each observation session with photographs, and using ImageJ to quantify petal discoloration.

5.3.2 Camera Position

Camera settings and the default camera position were standardised across observation sessions. Images were taken from directly above whenever possible. When the bent neck caused the flower head to droop, the angle was adjusted to capture the flower face directly, ensuring that petal-related criteria remained clearly visible regardless of stem orientation.

5.4 Nuisance Sources and Strategies

Table 5.4.1 *Summary of Nuisance Sources and Strategies*

Nuisance Source	Strategy
Temperature & Humidity	Monitor
Microbial Contamination	Control(Standardise)
Stem cutting variability	Control(Standardise)
Location	Block
Vase position within location	Randomise
Light exposure within location	Control
Vase/container variability	Control(Standardise)
Observer subjectivity	Control(Standardise)
Camera Position	Control(Standardise)

6. Independence and Confounding

6.1 Independence of Observations

The experimental unit was a single cut rose placed individually in a vase, with each vase receiving its own 120 mL allocation of the assigned treatment solution. Although solutions were prepared in larger batches for logistical convenience (§3.2), they were dispensed into individual vases before the roses were placed in them. Thus, no vase shared a treatment solution or remained connected to a common liquid source during the experiment.

Each vase was also observed and recorded separately at every observation session. Because the roses were physically separated, received individual solution allocations, and were measured independently, there was no direct mechanism by which the condition of one experimental unit could affect the condition of another. This design supports treating the experimental units as independent for the ANOVA model, with shared environmental sources of variation addressed separately through blocking and standardisation.

6.2 Potential Confounds and Resolution

Potential confounding by location was addressed through the blocking design described in §4.1 and §5.2.2. Because all four treatment combinations were represented equally within each location, treatment effects were not systematically aligned with location differences such as light exposure, air circulation, or room temperature. Location-related variation could therefore be accounted for as a block effect in the analysis.

One structural consequence of this design is that observer identity coincided with block: as noted in §5.3.1, each observer was responsible for a single location throughout the experiment. However, this structure does not create a treatment confound, because all four treatment combinations were assessed within each location and therefore by each observer. In addition, the use of standardised VBN criteria, photographic records, and ImageJ-based checks reduced the likelihood that observer subjectivity differentially affected the treatment comparisons.

Beyond this, the two factors of interest are not confounded with each other. Because the experiment used a full 2×2 factorial design, all four combinations of Sugar (with/without) and Vinegar (with/without) were represented, allowing the main effect of each factor and their interaction to be estimated independently of one another.

Procedural aspects of the experiment could, in principle, introduce additional confounds if applied unevenly across treatments. The 48-hour vase-solution replacement schedule, adopted to limit microbial contamination, was applied identically to all four treatment groups, and therefore cannot have differentially favoured any one condition over another.

7. Sample Size and Power Analysis

7.1 Power Analysis

According to De Silva et al. (2013), Extending the Vase Life of Gerbera (Gerbera hybrida) Cut Flowers Using Chemical Preservative Solutions, the average vase lives for distilled water (T2), vinegar treatment (T3), and sugar with vinegar treatment (T5) were 10.67, 12.67, and 14.33 days, respectively (SD = 0.57).

As data for all four treatment conditions were not available from a single source, the effect size was estimated from the three most comparable treatments reported by De Silva et al. (2013). This study, however, differs from the present experiment in two respects: it was conducted on Gerbera hybrida rather than Rosa hybrida, and the solutions were prepared using distilled water rather than tap water. In the absence of a more directly comparable source, De Silva et al. (2013) was selected as the closest available approximation.

To estimate Cohen's f from these three group means, the grand mean was first calculated as (10.67 + 12.67 + 14.33) / 3 = 12.557. The standard deviation of the group means was then computed as $\sqrt{[((10.67 - 12.557)^2 + (12.67 - 12.557)^2 + (14.33 - 12.557)^2) / 3]} = 1.496$. Dividing by the pooled SD yields Cohen's f = 1.496 / 0.57 = 2.625, indicating a large effect of water treatment type on vase life .This f value is used as an estimate of the effect size that water type is expected to have on vase life, applied to the present study's 2^2 factorial structure (k = 4) for the purpose of power calculation.

Unlike De Silva et al. (2013), which was conducted in a fully controlled environment, the present study is designed to assess the effects of water type in typical daily life conditions. Because complete environmental control is not feasible, greater error variance and an increased standard deviation are anticipated. Four scenarios were therefore constructed by incrementally increasing the assumed standard deviation to reflect possible increases in error variance under less controlled conditions.

Table 7.1.1 *Standard Deviation Scenarios and Corresponding Power at n = 8*

Scenario	SD	σ^2	Cohen's (f)	Replications	Power at n=8
Original SD	0.57	0.325	2.625	-	~1.00
2× SD	1.14	1.300	1.312	3	~1.00
3× SD	1.71	2.924	0.875	5	0.9818
4× SD	2.28	5.198	0.656	8	0.8390

Based on this analysis, n = 8 replications were selected, corresponding to an experimental design of at two locations, with 4 replications per at each location. To maintain an equal number of replications across locations,an even number of replications was required; 8 was therefore adopted as the most practical value. Even under the most conservative scenario (4×SD), n = 8 replications are expected to yield a statistical power of $1-\beta = 0.839$ at $\alpha = 0.05$, exceeding the conventional threshold of 0.80.

7.2 Practical Justification for the Effect Size

Beyond statistical power, the expected treatment differences were also evaluated in terms of practical meaningfulness from a consumer-choice perspective. De Silva et al. (2013) reported a 1.66-day difference in vase life between the vinegar-alone and sugar-plus-vinegar treatments, and a 3.66-day difference between the control and sugar-plus-vinegar treatments.

Rihn et al. (2014) provide choice-experiment evidence that differences of several days in expected vase life are economically meaningful to consumers. According to their study of single-species rose arrangements, consumers were willing to pay a premium for arrangements with an 8- to 10-day vase life relative to a 5- to 7-day baseline, and an even larger premium for arrangements with an 11- to 14-day vase life. Since these vase-life categories differ by approximately three days, the 3.66-day improvement reported by De Silva et al. (2013) can reasonably be interpreted as practically meaningful from a consumer perspective.

This supports the use of the De Silva et al. (2013) effect-size estimate as a target for the power analysis, indicating that the design was powered to detect a difference that is meaningful in practical as well as statistical terms.

8. Ethics and Safety

The experiment involved only the two team members, who observed and recorded the conditions of the roses. No external participants were involved, and no materials were consumed. The experiment posed minimal physical risk, as it involved only cut roses, water, sugar, and diluted vinegar. Therefore, no additional ethics or safety measures were required beyond ordinary care in handling the materials.

9. Execution

9.1 Plan and execution date

The table below summarizes the planned dates and actual execution dates, and indicates whether all rose observations were carried out until completion.

Table 9.1.1 *Dates of Planned and Actual*

Planned start date	29 May
Actual start date	29 May
Planned end date	7 June
Actual end date	12 June
Total runs completed	32
Runs excluded	None

9.2 Protocol deviations

The table below lists the pre-experiment plan, actual implementation, and reasons for changes made during the experiment.

Table 9.2.1 *Comparison of Planned and Actual Experiment plans*

Planned	Actual	Reason
Ambient temperature of approximately 23–25°C	Temperature range of approximately 25–28°C	Ambient temperature fluctuations
100mL per vase	120mL per vase	To ensure the rose stem is sufficiently submerged in water
Vase placed at the center of a full A4 sheet	Vase placed at the center of an A4 sheet cut in half	Insufficient experimental space
7 termination criteria selected from the VBN Rosa Evaluation Card	Exclude Leaf Shedding (Abscission) from termination criteria	The initial number of leaves for each rose is different
> 5 petals clearly (more than half) brown	> 50% of total petal surface area	The rose is not in a state where the entire surface area of the petals is visible.
1/100s shutter speed	1/10s shutter speed	Appears dark in indoor environments
The experiment ends on 7 June	Ended on 12 June	Roses lived longer than expected

9.3 Block Setup and Experimental Conditions

Two blocks were set up to accommodate a total of 32 experimental units, and 16 experimental units were placed in each block. Both blocks were conducted with direct sunlight blocked, and during the experiment, the temperature was maintained in the range of approximately 25–28°C and the humidity in the range of approximately 50–60%. Temperature and humidity were recorded along with the rose condition measurements.

9.4 Experiment execution

9.4.1 Rose Preparation Process

All roses were purchased at Yangjae-dong Flower Market around 8 a.m. on May 29. Light pink roses were selected to observe the yellowing of the petals. The thorns were removed from all the roses, and the stems were cut diagonally to a length of 18-20 cm. Additionally, any leaves below the vase rim or in poor condition were removed. The prepared roses were randomly placed into vases in Block 1 and Block 2 around 11:30 AM.

9.4.2 Vase Preparation Process

The 16 vases in each block were numbered from 1 to 16. All vases were the same model.

- Vase 1-4 : Tap water
- Vase 5-8 : 2% Sugar water
- Vase 9-12 : 0.6% White Vinegar water
- Vase 13-16 : contained 2% Sugar + 0.6% White Vinegar water

The original plan was to put 100mL of solution in each vase, but 120mL was put so that the rose stems would be sufficiently submerged in water. The solution was first replaced at 8:00 PM on May 31st and replaced every 48 hours. Roses whose vase life had ended were removed.

A Double A A4 sheet was cut at the midpoint of its long side to reduce its surface area by half, and a vase was placed in the center. Originally, the plan was to place the vase in the center of the A4 paper, but due to insufficient experimental space, half-sized paper was used. The vases were positioned randomly using R with a fixed random seed (set.seed([99])).

9.4.3 Measurement and Termination Process

Roses were observed at 8 AM and 8 PM, and the termination was determined based on six criteria selected from the VBN Rosa Evaluation Card. The six termination criteria are mentioned in §3.4.1.

The original plan had seven termination criteria, including Leaf Shedding (Abscission). However, since the initial number of leaves varied between roses, using Leaf Shedding (Abscission) as a termination criterion would have resulted in inconsistent comparisons across samples. Therefore, this criterion was excluded.

The Brown Flower criterion was changed from the condition “more than 5 petals clearly (more than half) brown” to “more than 50% of total leaf surface area.” Since the roses were not in a state where the entire surface area of the petals was visible, it was difficult to use the existing standard. Therefore, the procedure was carried out by applying the condition regarding leaf surface area, which is the same factor for yellowing, to the petals.

9.4.4 Photographic Conditions

Camera settings are mentioned in §3.4.2. The original plan was to set the shutter speed to 1/100s, but it was modified because it came out dark under artificial light. Additionally, A4 paper from Double A was used as the background.

9.5 Spreadsheet Description

The attached spreadsheet organizes the termination data for roses and consists of 8 columns and 33 rows. The data values recorded in each column are as follows.

- row A - ID
  Identification number for each of the 32 Vases. The IDs of vases 1-16 in Block 1 were recorded as 1-16 in order, and the IDs of vases 1-16 in Block 2 were recorded as 17-32 in order.

- row B - Block
  Location where each experiment was conducted. Distinguished as 1 and 2.

- row C - Vase
  Numbered from 1 to 16 by the Vase number in each Block

- row D - Water Type
  Types of solution in each vase
  : 1 - Tap Water / 2 - 2% Sugar Water / 3 - 0.6% Vinegar / 4 - 2% Sugar + 0.6% Vinegar Water

- row E - Freshness
  Record the score of the completed date. The date score starts at 0 on May 29 at 08:00 and increases by 0.5 every 12 hours.

- row F - Reason
  Reason for the decision to terminate
  : 1 - Bent Neck / 2 - Petal Discolouration / 3 - Yellow Foliage

- row G - Temperature
  Temperature on the terminated date

- row H - Humidity
  Humidity on the date of termination

10. Analysis

10.1 Statistical Model

The response variable, vase life, is measured in 0.5-day increments. Although technically discrete, the range of possible values is sufficient to treat it as approximately continuous, and two-way ANOVA is used as the primary analysis method, consistent with prior postharvest vase life research.

$$y_{ijkl} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma_k + \varepsilon_{ijkl}$$

- μ: the overall mean
- α_i : Main effect of Sugar , i = 1: Without Sugar, i = 2 With Sugar
- β_j : Main effect of Vinegar , j = 1: Without Vinegar, j = 2 With Vinegar
- $(\alpha\beta)_{ij}$: The interaction between the two factors
- γ_k: The block effect of location, k = 1, 2
- ε_{ijkl}: The random error term, assumed to be independently and identically distributed as $N(0, \sigma^2)$.

As mentioned in §5.2.2, Location is included as a fixed blocking factor.

Three hypotheses are tested through this experiment:

- H_0: $\alpha_1 = \alpha_2 = 0$ (Addition of Sugar has no effect on vase life) H_1: $\alpha_i \neq 0$ for at least one i
- H_0: $\beta_1 = \beta_2 = 0$ (Addition of Vinegar has no effect on vase life) H_1: $\beta_j \neq 0$ for at least one j
- H_0: $(\alpha\beta)_{ij} = 0$ for all i, j (no interaction between Sugar and Vinegar) H_1: $(\alpha\beta)_{ij} \neq 0$ for at least one i, j

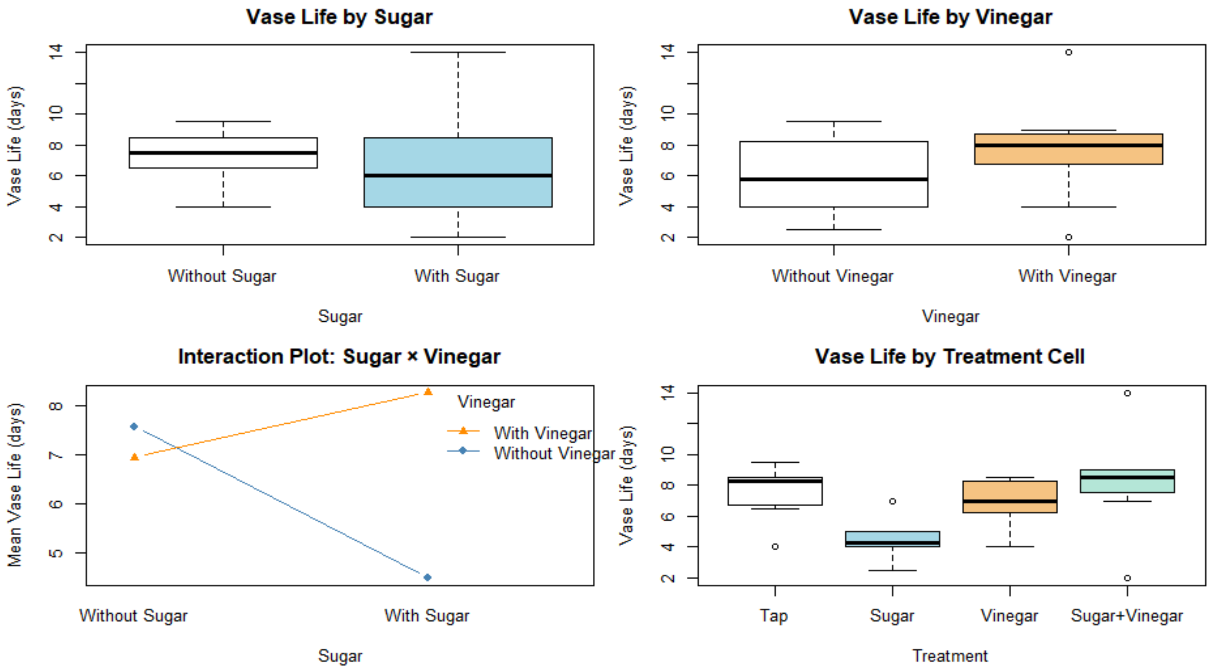
The interaction term is tested first at $\alpha = 0.05$. If the interaction is significant, main effects are not interpreted independently; instead, simple effects of Sugar within each level of Vinegar (and vice versa) are examined. If the interaction is not significant, the two main effects are interpreted independently.

10.2 Exploratory Data Analysis

Table 10.2.1 *Summary Table of treatments*

Treatment	Mean (days)	SD (days)	N
Tap	7.56	1.72	8
Sugar	4.50	1.28	8
Vinegar	6.94	1.50	8
Sugar + Vinegar	8.25	3.28	8

Figure 10.2.2 *Exploratory plots of vase life by treatment factor.*



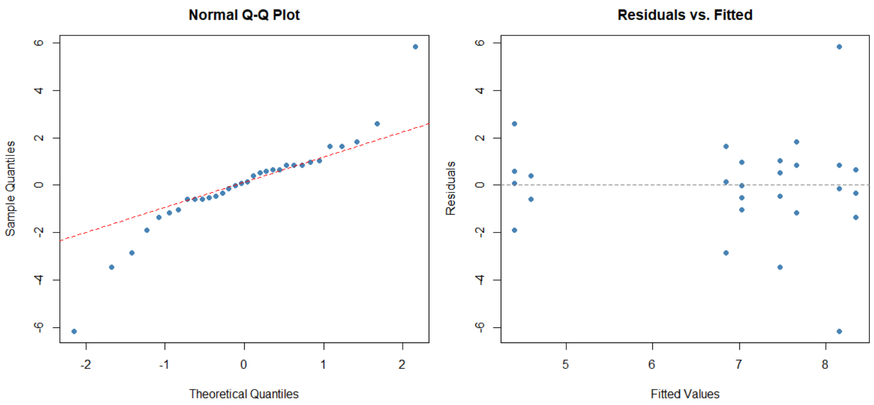
The upper plots show the marginal distributions of vase life by Sugar and Vinegar separately. Vase life appears slightly longer on average without Sugar than with Sugar. Conversely, vase life appears slightly longer with Vinegar than without Vinegar. However, because the interaction plot suggests a possible non-additive pattern, these marginal comparisons should be interpreted with caution.

The interaction plot in the lower-left panel shows that the two lines cross, indicating that the apparent effect of Sugar differs depending on whether Vinegar is present. Specifically, Sugar alone appears to shorten vase life relative to Tap water, whereas Sugar combined with Vinegar appears to extend it. This pattern of non-parallel, crossing lines suggests a Sugar × Vinegar interaction; accordingly, the interaction term is examined first in the formal analysis (§10.4).

The four-cell boxplot in the lower-right panel further illustrates this pattern while also showing the within-treatment spread. Several visually apparent potential outliers are present across cells, including one each in the Tap and Sugar groups and two in the Sugar + Vinegar group. This suggests notable variability within some treatment combinations, which is examined further in the model diagnostics (§10.3).

10.3 Model Diagnostics

Figure 10.3.1 *Diagnostic Plots*



10.3.1 Assessment of Assumptions

Normality

The Q-Q plot follows the reference line reasonably well in the central region, but shows noticeable departures at both tails. These departures appear to be driven by a small number of extreme observations, including the two extreme residuals in the Sugar + Vinegar cell. Therefore, the normality assumption is considered approximately reasonable, although the tail departures should be noted.

Constant Variance

The residuals vs. fitted plot shows no clear systematic fanning or funnel pattern across fitted values. Although the spread is influenced by a few extreme observations, especially at the higher fitted values, there is no strong visual evidence of nonconstant variance. Thus, the constant variance assumption is considered reasonable.

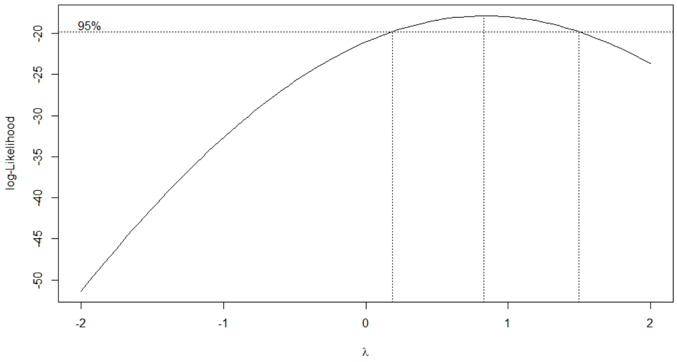
Outliers

Externally Studentized residuals were computed and compared against a Bonferroni-corrected critical value of 3.52 ($\alpha = 0.05$, $n = 32$). Two observations were identified as outliers: ID 29 ($t = 3.57$) and ID 30 ($t = -3.86$), both belonging to the Sugar+Vinegar cell in Block 2, with vase life values of 14 days and 2 days respectively.

As no protocol violations or measurement errors were recorded for these observations, they were not removed outright. Instead, following the recommendation of Oehlert (2010), the analysis was conducted both with and without these observations, and the results are compared in §10.4.

10.3.2 Transformation

Figure 10.3.2 *Box -Cox Plot*



A Box-Cox analysis was conducted to assess whether a transformation of the response was necessary. The estimated optimal λ was 0.83, which is relatively close to 1. Furthermore, the 95% confidence interval for λ includes 1. This suggests that transforming the response would not substantially improve the model. Therefore, the original response scale was maintained.

10.4 ANOVA Table

Figure 10.4.1 *ANOVA for Full Data*

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sugar	1	6.12	6.12	1.343	0.25669
Vinegar	1	19.53	19.53	4.282	0.04821 *
Block	1	0.28	0.28	0.062	0.80577
Sugar:Vinegar	1	38.28	38.28	8.393	0.00738 **
Residuals	27	123.16	4.56		

Figure 10.4.2 *ANOVA for Filtered Data*

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sugar	1	9.15	9.15	4.479	0.044444 *
Vinegar	1	15.25	15.25	7.462	0.011390 *
Block	1	0.87	0.87	0.427	0.519416
Sugar:Vinegar	1	36.01	36.01	17.622	0.000298 ***
Residuals	25	51.09	2.04		

The blocked two-way ANOVA showed a statistically significant Sugar × Vinegar interaction in the full-data analysis, ($F(1, 27) = 8.393$, $p = 0.007$). The same interaction remained significant after excluding the two visually apparent extreme observations, ($F(1, 25) = 17.622$, $p < 0.001$). In addition, the interaction sum of squares changed only slightly, from 38.28 in the full data to 36.01 in the filtered data. This suggests that the Sugar × Vinegar interaction was not solely driven by the potential outliers.

The significant interaction indicates that the effect of Sugar on vase life was not constant across Vinegar levels. Based on the cell means and interaction plot, Sugar alone appeared to shorten vase life relative to Tap water, whereas Sugar combined with Vinegar appeared to extend vase life. Because the interaction term dominated the treatment structure, the treatment combinations are examined at the cell level in the follow-up analysis (§10.5)

10.5 Main Analysis

To identify which treatment combinations differ significantly, Tukey's HSD is applied across all four cells, controlling the familywise error rate simultaneously across all pairwise comparisons. The simple effects of Sugar at each level of Vinegar are then examined to characterize the direction of the interaction.

10.5.1 All Pairwise Comparisons: Tukey's HSD

Tukey's HSD was applied across all four treatment cells to control the familywise error rate at $\alpha = 0.05$ while providing simultaneous 95% confidence intervals for all pairwise differences.

Table 10.5.1.1 *Tukey's HSD for full data*

Comparison	Diff	lwr	upr	p adj
Sugar-Tap	-3.0625	-5.98478	-0.14022	0.03738 *
Vinegar-Tap	-0.6250	-3.54728	2.29728	0.93566
Sugar+Vinegar-Tap	0.6875	-2.23478	3.60978	0.91678
Vinegar-Sugar	2.4375	-0.48478	5.35978	0.12731
Sugar+Vinegar-Sugar	3.7500	0.82772	6.67228	0.00812 ***
Sugar+Vinegar-Vinegar	1.3125	-1.60978	4.23478	0.61431

In the full-data analysis, two of the six pairwise comparisons were statistically significant. Sugar differed significantly from Tap water, indicating that Sugar alone shortened vase life. Sugar+Vinegar differed significantly from Sugar,indicating that the addition of Vinegar reversed the negative effect of Sugar. The remaining four comparisons, including Vinegar vs. Tap and Sugar+Vinegar vs. Tap, were not statistically significant.

Table 10.5.1.2 *Letter Display for full data*

Treatment	**Group**
Sugar + Vinegar	a
Tap	a
Vinegar	ab
Sugar	b

Figure 10.5.1.3 *Underline Diagram for full data*

The underline diagram shows that Sugar formed a separate lower-mean group from Tap and Sugar+Vinegar, while Vinegar overlapped with both groups and was not significantly different from any other treatment.

Table 10.5.1.4 *Tukey's HSD for filtered data*

Comparison	**Diff**	**lwr**	**upr**	**p adj**
Sugar-Tap	-3.0625000	-5.0284930	-1.096507	0.0012777 ***
Vinegar-Tap	-0.6250000	-2.5909930	1.340993	0.8179352
Sugar+Vinegar-Tap	0.7708333	-1.3526818	2.894348	0.7516262
Vinegar-Sugar	2.4375000	0.4715070	4.403493	0.0111074 **
Sugar+Vinegar-Sugar	3.8333333	1.7098182	5.956848	0.0002255 ***
Sugar+Vinegar-Vinegar	1.3958333	-0.7276818	3.519348	0.2932623

After excluding the two extreme observations (§10.3), the same two comparisons remained significant, with narrower confidence intervals reflecting the reduction in MSE from 4.56 to 2.04.

One additional comparison, Vinegar vs. Sugar reached significance in the filtered analysis. This suggests that the full-data analysis was somewhat underpowered for detecting this difference.

Table 10.5.1.5 *Letter Display for filtered data*

Treatment	**Group**
Sugar+Vinegar	a
Tap	a
Vinegar	a
Sugar	b

Figure 10.5.1.6 *Underline Diagram for filtered data*

In the filtered analysis, Sugar was significantly different from all three remaining treatments; Tap, Vinegar, and Sugar+Vinegar formed a single non-significant group, with Sugar standing apart as the only distinct group.

In conclusion, across both analyses, Sugar+Vinegar produced the numerically highest mean vase life but was not statistically distinguishable from Tap or Vinegar; the most consistent finding is that Sugar alone significantly shortened vase life relative to all other treatments.

10.5.2 Simple Effects of Sugar at Each Level of Vinegar

Table 10.5.2 *Table for Simple Effect Analysis*

Simple Effect of	**Tukey HSD Row**	**Diff (Full/Filtered)**	**p (Full)**	**p (Filtered)**
Sugar (without Vinegar)	Sugar − Tap	−3.06	0.037 *	0.001 ***
Sugar (with Vinegar)	Sugar+Vinegar − Vinegar	+1.31 / +1.40	0.614	0.293
Vinegar (without Sugar)	Vinegar − Tap	−0.625	0.936	0.818
Vinegar (with Sugar)	Sugar+Vinegar − Sugar	+3.75 / +3.83	0.008 **	0.0002 ***

Without Vinegar, Sugar significantly reduced mean vase life relative to Tap in both the full and filtered models. With Vinegar present, however, adding Sugar increased mean vase life relative to Vinegar alone, although this difference was not statistically significant in either model.

On the other hand, Vinegar alone had no significant effect relative to Tap in either model. However, when combined with Sugar, Vinegar significantly increased vase life relative to Sugar alone in both models.

These comparisons support the crossing pattern observed in §10.2. Sugar alone was associated with a shorter vase life, but this negative effect was not observed when Vinegar was also present. Similarly, Vinegar had little effect in the absence of Sugar but substantially improved vase life when added to Sugar. The treatment response is therefore best interpreted as a Sugar × Vinegar interaction rather than as two separate marginal main effects.

11. Conclusions

11.1 Summary of Findings

This experiment investigated the effects of Sugar and Vinegar as vase solution additives on the vase life of cut roses using a blocked 2×2 factorial design.

A significant Sugar × Vinegar interaction was identified, indicating that the effects of the two additives cannot be interpreted independently. The simple effects analysis showed that Sugar shortened vase life when used without Vinegar, whereas this negative effect was not observed when Vinegar was also present. Conversely, Vinegar had little effect relative to Tap when used alone, but significantly increased vase life when added to Sugar. Although Sugar + Vinegar produced the numerically highest mean vase life in the full dataset, it was not statistically distinguishable from Tap water or Vinegar alone.

The clearest finding is that Sugar without Vinegar produced the shortest vase life, with statistically significant reductions relative to Tap and Sugar + Vinegar in both the full and filtered analyses.

11.2 Justification of Conclusions

These findings suggest that Sugar without Vinegar shortened vase life and that the treatment response depended on the Sugar–Vinegar combination. A study on *Rosa hybrida* using 2% sucrose—the same species and sugar concentration as the present experiment—found that sucrose treatment alone shortened vase life compared with tap water, attributing this pattern to microbial proliferation in the sucrose-only condition (Halevy and Mayak, 1981, as cited in Lee et al.).

This provides a plausible biological explanation for the present results. Sucrose without a germicide can promote bacterial growth, leading to xylem blockage and reduced water uptake, which may shorten vase life. The significant advantage of Sugar + Vinegar over Sugar alone in the present experiment is consistent with vinegar acting as an acidic antimicrobial component, although microbial growth was not directly measured.

11.3 Limitations

This study is subject to several limitations. With only eight replications per treatment cell and two blocks, the statistical power of the design depended heavily on the assumed effect size, which was derived from a different species under controlled conditions. If the true effects in the present experiment were smaller than assumed, some real differences may not have reached statistical significance.

Moreover, in the filtered analysis, the removal of the two extreme observations further reduced the Sugar + Vinegar cell to six observations. Several pairwise comparisons, including Sugar + Vinegar versus Tap and Vinegar versus Tap, remained non-significant in both the full and filtered analyses. Therefore, it is not possible to determine whether these null results reflect a genuine absence of effect or insufficient replication. In addition, the effect size used in the power analysis was derived from experiments on *Gerbera hybrida* rather than *Rosa hybrida*, so the assumed effect size may not have been directly transferable to the present experiment.

A further limitation concerns the Sugar + Vinegar treatment cell, which exhibited markedly higher within-cell variability (SD = 3.28 days) than the remaining three cells (SD = 1.28–1.72 days). The two observations identified as outliers in §10.3 both belonged to this cell, and their opposing extremes, 14 days and 2 days, suggest that the combined treatment may have been sensitive to uncontrolled variation that was not identified during the experiment. This limits the precision of estimates involving the Sugar + Vinegar condition and indicates caution in interpreting the magnitude of the interaction effect, even though the interaction remained statistically significant in both the full and filtered analyses.

Finally, the generalisability of these findings is limited primarily by the fixed treatment concentrations used in the experiment. The study tested only one sugar concentration (2%) and one white vinegar concentration (0.6%), both drawn from a prior study. Therefore, the results should be interpreted as evidence about these specific treatment levels rather than as evidence about the optimal concentration of sugar or vinegar more generally. Different concentrations may produce different effects, and the observed Sugar × Vinegar interaction may not hold across other dosage combinations.

11.4 Directions for Future Research

Future studies should increase the number of replicates per treatment cell to improve statistical power and to better distinguish whether the non-significant differences among Tap, Vinegar, and Sugar + Vinegar reflect a genuine absence of effect or insufficient replication.

Future work should also extend the range of treatment concentrations beyond the 2% Sugar and 0.6% Vinegar levels tested in this study. A factorial design with multiple Sugar and Vinegar concentrations would allow researchers to examine dose-response relationships and identify potentially more effective combinations.

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Appendix A

ImageJ was used to objectively measure the yellowing of petals and leaves. The photographs attached in Appendix A were included when the yellowing of petals and leaves was the terminating factor in the roses.

The photos marked with yellow lines indicate the outlines of the yellowed areas. The numbers written at the bottom of each rose photo represent, in order of
: (Block number)_(Termination date)_(Termination time)_(Vase number).

Among the numbers written below ImageJ Analysis, 1 represents the entire petal, and 2 represents the yellowed petal. The four numbers written next to 1 and 2 represent Area, Mean, Min, and Max, respectively. The number next to the equal sign indicates the ratio of the yellowed petal area to the total petal area. If this number is higher than 0.5, it was recorded and saved in Appendix A.

Photo	ImageJ Analysis
 1_20260604_2000_12	1 3416269 116.457 27 197 2 1715387 108.805 50 163 = 0.5021
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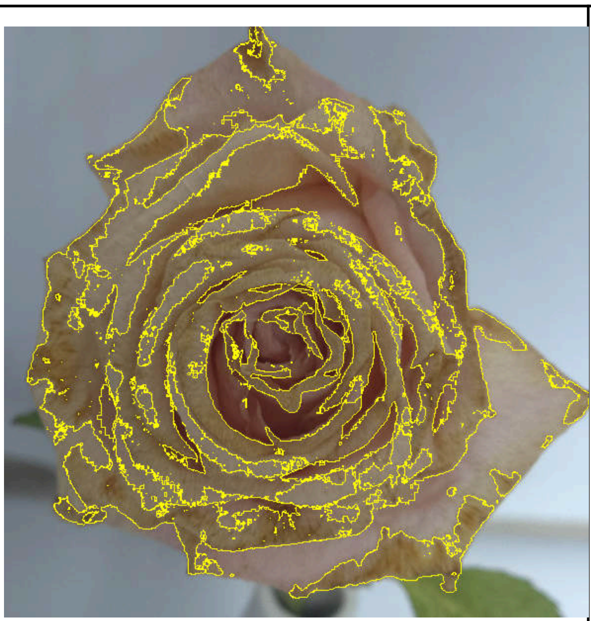
<tr><td>
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